

Community Detection in the European Football Transfer Market

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Abstract

The European football transfer market can be represented as a complex network in which clubs exchange players through transfer relationships. In this work, we investigate whether community detection algorithms can recover the underlying league structure of the European transfer market. Using transfer records from nine major European leagues, we construct weighted transfer networks and compare the performance of several community detection algorithms, including Louvain, Clauset–Newman–Moore, Label Propagation, Kernighan–Lin, and Girvan–Newman. The detected communities are evaluated against league membership using modularity, Normalized Mutual Information (NMI), and Adjusted Rand Index (ARI). We further analyse the robustness of the detected communities under edge removal, hub removal, and transfer-threshold perturbations, and study their evolution across multiple seasons. Our results show that domestic league membership is the dominant factor shaping community structure, with Louvain achieving the strongest agreement with the league-based ground truth. The identified communities remain relatively stable across seasons and robust to moderate network perturbations, highlighting the effectiveness of community detection methods for analysing football transfer networks.

1 Introduction

The transfer of football players is a real-world economic network: we can imagine clubs as nodes and the number of transfers between two clubs as a weighted edge.

Understanding the structure of football transfer networks is relevant for both sports analytics and network science. Transfer relationships

shape talent development pathways, influence the competitive balance between leagues, and reflect underlying economic and geographical connections between clubs. Identifying communities within these networks can therefore provide insights into how football markets are organised and how players circulate across Europe.

1.1 Research questions and central hypothesis

The European football transfer market naturally forms a complex network in which clubs exchange players across national and international boundaries. Since clubs belonging to the same domestic league are expected to trade players more frequently among themselves than with clubs from other countries, we hypothesize that community detection algorithms should largely recover the underlying league structure. However, cross-national transfer corridors, shared-ownership relationships, and cultural or geographical proximity effects may generate communities that do not perfectly align with league boundaries.

Based on this hypothesis, our study addresses the following research questions:

- **RQ1. To what extent can community detection algorithms recover the national league structure of the European football transfer network?**

We evaluate whether the communities identified from transfer interactions correspond to the ground-truth partition defined by domestic league membership.

- **RQ2. Do different community detection algorithms differ in their ability to recover the underlying league structure?**

We compare several algorithms, including Louvain, Greedy Modularity (Clauset-Newman-Moore), Label Propagation, Kernighan-Lin Bisection, and Girvan-Newman, using metrics such as modularity, number of detected communities, and agreement with the league-based ground truth.

- **RQ3. How does the community structure of the transfer network evolve over time?**

Using season-by-season transfer networks, we investigate the temporal stability of communities and assess whether similar communities can be identified and tracked across different football seasons.

- **RQ4. How robust are the detected communities to structural perturbations of the network?**

We examine the sensitivity of community assignments under random edge removal, iterative removal of high-degree hub clubs, and variations in the minimum transfer threshold used to construct the network.

2 Related Work

Recent work has increasingly applied network science methods to football transfer markets. Previous studies have analysed the structural properties of transfer networks, the role of central clubs, and the relationship between network position and sporting success [1–3]. Other works have explored community structures and trading patterns in football transfer networks at both national and global scales [4, 5].

Unlike previous studies that primarily focused on network centrality, team performance, or the description of transfer-market communities, our work provides a systematic comparison of community detection algorithms, evaluates their agreement with league-based ground truth, and analyses both the temporal evolution and robustness of the detected communities.

3 Data and network construction

3.1 Data loading and preprocessing

The raw datasets containing transfers information of different European national leagues were first explored to understand their structure and identify potential data quality issues. Several preprocessing steps were then applied before network construction, including the removal of duplicate transfer records, normalization of club names across seasons and leagues, and the treatment of clubs belonging to leagues outside the scope of the study. These steps were necessary to ensure consistency and reliability of the resulting transfer network.

A substantial portion of the overall project effort was devoted to data preprocessing and quality assurance, as ensuring consistent club identities across multiple leagues and seasons proved essential for the reliability of the subsequent network analysis. Further details regarding the preprocessing pipeline are provided in Appendix A.

3.2 Network model

After completing the preprocessing steps, we obtained a cleaned dataset containing, for each transfer, the source club, destination club, and season of the transaction. Additionally, a league mapping dictionary was created, assigning every club present in the dataset to its corresponding league.

To construct the transfer networks, the cleaned dataset can be filtered by season and transformed into a weighted undirected graph. Nodes represent football clubs, while edges represent transfer relationships between clubs. Edge weights correspond to the total number of player transfers exchanged between two clubs during the selected time period, regardless of transfer direction.

3.3 Ground truth definition

Since our main objective is to evaluate whether community detection algorithms recover the structure of the European football transfer market, a reference partition is required. We define the ground truth as the domestic league

membership of each club. Thus, clubs belonging to the same national league are assigned to the same ground-truth community. Clubs originating from leagues outside the nine leagues included in the study are assigned to a separate category labelled Other.

This ground-truth partition is subsequently used to compute Normalized Mutual Information (NMI) and Adjusted Rand Index (ARI) scores between detected communities and league membership. NMI quantifies how much information about the league partition is captured by the detected communities, whereas ARI measures how often pairs of clubs are assigned consistently to the same or different groups, correcting for random agreement. Higher values indicate stronger correspondence with the ground truth.

4 Basic network characterization

Before applying community detection algorithms, we first examine the structural properties of the transfer network. This provides an overview of the network topology and helps contextualize the community structures identified in subsequent sections. The chosen network to study and apply different algorithms on, consists on all the transfer records comprising from the 2020/2021 to the 2022/2023 season.

Table 1 summarizes the main characteristics of the network.

Table 1: Basic network statistics.	
Metric	Value
Number of nodes	1191
Number of edges	6058
Density	0.0085
Average clustering coefficient	0.156
Connected components	1

The network contains 1,191 clubs connected through 6,058 transfer relationships. The low density value (0.0085) indicates that only a small fraction of all possible club-to-club connections are observed (number of observed edges over maximum possible number of edges that could exist in the network), which is expected in a transfer market where most clubs

exchange players with only a limited subset of other clubs.

The average clustering coefficient (0.156) indicates the presence of meaningful local structure in the transfer market, as it is considerably higher (≈ 18 times) than the network density, which would be the expected average clustering coefficient for in a random graph. Thus, Clubs connected to the same club are relatively likely to also exchange players among themselves, suggesting the existence of localized transfer ecosystems that may be captured by community detection algorithms.

The network consists of a single connected component, meaning that every club is connected, directly or indirectly, through chains of transfer interactions. This property is particularly relevant for community detection, as it implies that the transfer market forms a single large interconnected system rather than several disconnected subnetworks.

Figure 2 in the appendix shows the degree distribution of the network.

The degree distribution is highly right-skewed. Most clubs have relatively few transfer connections, while a small number of clubs exhibit substantially higher degrees and act as hubs within the network. This heterogeneity suggests that transfer activity is not uniformly distributed across clubs, with a limited number of highly connected clubs playing a central role in linking different parts of the transfer market.

5 Community detection: algorithm comparison

After the network definition, we can now run all the different community detection algorithms. Those will find disjoint communities as every club is part of a single league.

5.1 Algorithms and motivation

We will briefly explain the algorithms we chose and why we chose them:

Louvain [6] The Louvain algorithm optimises modularity (Q), a measure that compares the observed number of intra-community connections with the number expected in a random network with the

same degree distribution. It operates through a two-phase iterative procedure. In the first phase, each node is greedily re-assigned to the neighbouring community that yields the largest modularity gain, until no further improvement is possible. In the second phase, each community is collapsed into a single super-node and the process repeats on the coarsened graph.

It is the standard for large networks due to its near-linear time complexity with network size n , $\mathcal{O}(n \log n)$ and consistently high modularity scores, and it will be our primary benchmark as we expect it to be the one that better reflects the ground truth.

Clauset–Newman–Moore (CNM) [7]

CNM is an agglomerative algorithm that starts with each node in its own community and repeatedly merges the pair of communities whose fusion produces the largest modularity increase. The process terminates when no merge is beneficial, yielding the partition that maximises Q along the resulting dendrogram.

Because modularity landscapes are known to contain many near-optimal solutions, running a second optimiser lets us assess whether the modularity peak found by Louvain is unique or whether structurally different partitions achieve similar scores.

Label Propagation (LPA) [8] LPA detects communities without optimising any global objective. Each node is initialised with a unique label and then iteratively adopts the label held by the majority of its neighbours until the labels stabilise. Communities emerge as groups of nodes sharing the same final label.

LPA does not optimise modularity and has no explicit null model; it discovers communities purely from local majority dynamics. We want to find out if some community structure are detectable without a modularity-based objective, even knowing that, because of how highly connected this network is, it might collapse in a single community.

Kernighan–Lin (KL) Bisection [9]

Kernighan–Lin is a graph-partitioning heuristic that divides the network into exactly two groups by iteratively swapping pairs of nodes between the groups to minimise the weighted cut, the total weight of edges crossing the partition boundary.

Because it is constrained to produce exactly two communities regardless of the underlying structure, it is used here as a sanity check rather than a general community detector: the resulting bisection reveals the dominant structural divide in the transfer network. (e.g. a geographic east–west or north–south split).

Girvan–Newman (GN) [10]

Girvan–Newman takes a divisive approach: it progressively removes the edge with the highest betweenness centrality (the edge traversed by the greatest number of shortest paths) and recomputes betweenness after each removal. Edges that bridge different communities carry the most shortest paths and are therefore removed first, gradually exposing the community structure as a dendrogram. The partition selected is the one that maximises Q along the hierarchy.

All the algorithms are already implemented on python libraries, so no manual implementation was needed. In particular, Louvain implementation can be found at [11] and all the other algorithms at [12]. There are no parameters to be tuned, and they are able to handle weighted networks. Every algorithm will be run 5 times, except GN that required almost 3 hours for a single run.

5.2 Discussion of null model

Only Louvain and CNM use modularity to measure the excess of intra-community edges relative to a null model. For this purpose we chose the configuration model, which is what would be expected by chance given each node’s degree. This is the most adapt one because our network is heterogeneous: a handful of hub clubs (e.g. Chelsea, Juventus, Manchester City, Benfica) participate in far more transfers than

peripheral clubs, so maintaining the degree of the nodes is fundamental.

5.3 Results

Scores obtained by every algorithm are showed in Figure 3. As we were expecting, Louvain and CNM outperformed the other three algorithms on reflecting ground truth and LPA collapsed into two communities. GN was the most unpredictable and indeed it created 15 communities. An overall distribution of leagues into communities can be seen in Figure 6, where clubs which don't belong to any of the major leagues are not considered. For clarity, since we run the algorithms 5 times, Figure 6 reflects the run with the best outcome based on the metrics. Let's now deep dive into every algorithm's outcome.

Louvain Louvain successfully clustered all the 9 leagues in 8 communities, with one being the union of the Premier League and Championship. This is a plausible result, as both competitions belong to the same national football structure and exhibit strong transfer interactions, reflecting their geographical proximity and shared institutional environment. The performance of this algorithm in recovering the league structures is surprisingly strong, obtaining an average purity of 0.904 in the detected communities, which means that on average, the dominant league in each community represents 90% of the total different leagues that the community encompasses. Analyzing the clubs in the communities coming from Other leagues, we can find that community number 1 (*C1*), mainly composed by portuguese clubs, has also many clubs from Brasil, highlighting the cultural and language influence. Same happened for *C3*, where clubs from France's colonies are clustered together with Ligue 1 clubs. Dutch and Belgium's clubs ended up in the same community, reflecting a well-known market history between these two leagues. Figure 4 in the appendix shows the coloring of detected communities in the network.

Clauset–Newman–Moore (CNM) CNM

recovers the same broad structure as Louvain but produces only 7 communities instead of 8, merging the smaller national clusters due to the resolution limit. Specifically, the German/Austrian/Swiss cluster and the Dutch/Belgian cluster are fused into *C2*, as neither is large enough to be detected independently under CNM's greedy merging strategy. A detailed list of the clubs contained in this community for this algorithm are shown in Figure 5 in the appendix. This results are reflected in the average purity of the communities obtained by the algorithm, which is of 75%. This shows the tendency of richer clubs to get talents from clubs outside their country, which couldn't be noticed in Louvain's communities because Germany and Netherlands were separated and so english clubs' influence was stronger. The four larger ecosystems identified by Louvain survive intact, confirming that they represent the strongest structural signals in the network.

Label Propagation (LPA) Due to the high number of connections in the network, LPA failed to distinguish different communities. Exception made for a little group of small clubs from Liga NOS, which were clustered together with small clubs from Brasil and Venezuela, indicating that they probably have most of their market between each others.

Kernighan–Lin (KL) Bisection As a recall, we used this algorithm to see where the main division would have been in case of bisection. The outcome is basically a split between those countries that are directly bathed by the Atlantic Ocean (England, Portugal, Spain and France) and those who are not (Italy, Germany, Netherlands, Russia). This indicates that if we have to split the European market in half based on where trades happen the most, we can put center-eastern Europe on a side and western Europe on the other.

Girvan–Newman (GN) This is by far the most curious one: *C0* contains all the leagues, but for the first time we see some russian club in it, those that have some

trade outside the east of Europe; anyway, russian league and east-european clubs form a separated cluster (*C1*). Then, from *C2* to *C12* and *C14*, portuguese and related clubs present in Louvain’s *C0* were splitted into smaller communities. Some group Portuguese lower-division clubs with their Brazilian counterparts in specific regions; others reveal a distinct Portuguese-Asian pipeline, where Brazilian and Portuguese intermediaries systematically broker transfers toward Japanese, Chinese, Iranian, and Malaysian clubs, a flow that is structurally strong enough to form its own community. Finally, *C12* isolates clubs from a specific Portuguese island, Madeira, whose local scouting network operates independently enough from the mainland to produce a detectable structural signature in the transfer graph.

Given its superior overall performance in recovering the league-based ground truth, as well as its high modularity scores, Louvain was selected for the robustness and temporal analyses presented in the following sections. Focusing on a single algorithm also allows for a more interpretable assessment of how community structures respond to network perturbations and evolve over time.

6 Robustness and sensitivity analysis

Outcomes of the experiments can be seen in Figure 7, where Edge Removal: Modularity shows how the modularity score Q changes as an increasing fraction of edges is randomly deleted.

Edge Removal: Partition Stability tracks NMI and ARI against the baseline partition (the full graph) as edges are removed.

Edge Removal: Community Count shows how the number of detected communities evolves with edge removal.

Hub Removal: Metrics Evolution shows modularity, NMI, and ARI as the top- k highest-degree clubs are progressively removed. It reveals how much the partition depends on the most connected clubs.

Hub Removal: Community Count tracks how many communities Louvain detects after each hub is removed. As hubs bridge multiple communities, removing them can either split or merge communities depending on which connections they maintained.

Threshold Sweep: Metrics vs Min Weight shows how modularity, NMI (against league ground truth), and ARI change as only edges with weight above a given threshold are retained, alongside the number of surviving nodes (right axis).

6.1 Edge removal

To test sensitivity to missing data, we randomly removed increasing fractions of edges (from 2% to 50%) and re-ran Louvain 20 times per condition, measuring NMI and ARI against the baseline partition. The community structure degrades gradually but remains stable: even at 10% edge removal, NMI stays above 0.80 and ARI above 0.79, indicating that the broad partition is recoverable under moderate noise. Modularity is virtually unaffected across all conditions (ranging from 0.500 to 0.511), confirming that the algorithm finds equivalently good solutions on perturbed graphs. The number of detected communities remains stable around 7–8 throughout, suggesting that the macro-level structure is not an artifact of individual edges. Only at 50% removal does agreement drop to NMI 0.57, at which point the graph has lost structural coherence rather than just noise.

6.2 Edge removal

To test sensitivity to missing data, we randomly removed increasing fractions of edges (from 2% to 50%) and re-ran Louvain 20 times per condition, measuring NMI and ARI against the baseline partition (771 nodes, 8 communities, $Q = 0.489$). Community structure degrades gradually but remains recoverable under moderate noise: at 2% removal, NMI is already 0.80 and ARI 0.78, indicating that even small perturbations cause some community reassignment. At 10% removal NMI drops to 0.69 and ARI to 0.65, while modularity remains virtually unaffected (0.496 ± 0.007). The number of detected communities grows

slightly with noise (from 8 to 9 by 30%), reflecting minor boundary fragmentation rather than a structural collapse. Only at 50% removal does agreement fall substantially (NMI 0.46, ARI 0.34), with the community count inflating to 13 and modularity rising to 0.558, a sign that the algorithm is partitioning an incoherent residual graph rather than recovering meaningful structure.

6.3 Iterative hub removal

We iteratively removed the top- k clubs by weighted degree and re-ran Louvain at each step, comparing partitions on the remaining nodes. The top hub is Sparta (weighted degree 70), followed by Benfica (59) and Sporting (56). Removing only the top hub immediately drops NMI to 0.80 and ARI to 0.79 ($k=1$), confirming that high-degree nodes are disproportionately informative for the partition. Interestingly, NMI partially stabilises between $k=3$ and $k=10$ (NMI ≈ 0.75 – 0.76), suggesting that once the most disruptive connector is gone the remaining structure is relatively coherent. Modularity increases monotonically with hub removal (0.489 at baseline to 0.616 at $k=50$). This is a well-known artefact of the modularity measure: high-degree nodes connect many communities at once, and their presence makes it harder for the algorithm to draw sharp boundaries between groups. Once they are removed, the remaining clubs interact more exclusively within their own cluster, making the partition appear cleaner even though the underlying structure has not fundamentally changed. Even after removing the top 50 nodes (reducing the graph to 598 nodes), NMI remains at 0.68 and ARI at 0.63, demonstrating that the macro-level partition is not dependent on any single club.

6.4 Edge-weight threshold sweep

We filtered the network to retain only edges with weight above increasing thresholds (1 to 5), removing low-frequency transfers. The network shrinks sharply: from 771 nodes and 2540 edges at threshold 1, to 278 nodes and 392 edges at threshold 2, to only 4 nodes at threshold 5. Modularity rises steeply from 0.489 to 0.922 as the network is pruned, since sur-

viving high-frequency connections form tight, well-separated cliques. NMI against the league ground truth peaks at threshold 4 (NMI 0.69), while ARI remains low throughout (max 0.57 at threshold 5, on only 4 nodes). This suggests that the most repeatedly traded clubs do align with league identity, but the sparse high-threshold graph is too small to constitute a meaningful community structure. The experiment confirms that the full network contains substantial noise from one-off transfers, and that high-frequency transfer relationships are the primary driver of league-coherent community structure.

Table 2: Louvain community detection results across five seasons.

Season	Comm.	Mod.	NMI	ARI	Purity
2018/19	8	0.550	0.790	0.719	0.879
2019/20	8	0.528	0.714	0.637	0.813
2020/21	7	0.529	0.661	0.554	0.756
2021/22	8	0.537	0.712	0.603	0.792
2022/23	8	0.489	0.757	0.693	0.855

Table 2 summarizes the results obtained for the five seasonal networks. Across all seasons, Louvain consistently identifies between seven and eight communities, suggesting that the overall community structure of the European transfer market remains relatively stable over time.

The modularity values range between 0.49 and 0.55, indicating a clear community structure in every season. This suggests that transfer interactions are not randomly distributed across clubs, but instead exhibit persistent clustering patterns. Similarly, the average community purity remains relatively high throughout the observation period, with values between 0.76 and 0.88, indicating that most detected communities are dominated by a single league.

Agreement with the ground-truth league partition is also consistently strong. NMI values range from 0.66 to 0.79, while ARI values range from 0.55 to 0.72. Although some fluctuations can be observed, the results indicate that domestic league membership remains one of the primary factors shaping the community structure of the transfer market.

A qualitative inspection of the detected communities further reveals that several league-

centered communities persist throughout the entire period. Communities dominated by La Liga, Serie A, Ligue 1, the Premier League, and Liga NOS are repeatedly recovered across seasons, suggesting that these structures are stable over time. At the same time, a small number of mixed communities appear in certain seasons, reflecting cross-league transfer relationships and broader transfer ecosystems that cannot be explained solely by domestic league boundaries.

An interesting example occurs in the 2020/2021 season, where Louvain identifies only seven communities instead of eight. In this case, clubs from the German Bundesliga and the Dutch Eredivisie are grouped into a single community rather than forming separate league-centered clusters. This suggests that transfer interactions between clubs from these two leagues were particularly strong during that season. While the network data alone does not allow us to determine the exact causes, geographical proximity, cultural similarities, and established transfer pathways between German and Dutch clubs may contribute to this behaviour.

7 Conclusion

In this work, we studied the European football transfer market as a weighted network of clubs and investigated its community structure using several community detection algorithms.

Our results show that the transfer market exhibits a clear and persistent community structure. Across all experiments, domestic league membership emerged as the dominant organizing principle of the network, confirming our initial hypothesis that clubs belonging to the same league tend to exchange players more frequently among themselves than with clubs from other countries.

Among the evaluated algorithms, Louvain achieved the best overall performance, obtaining the highest modularity, NMI, ARI, and community purity scores. The detected communities closely matched the ground-truth league partition, while also revealing meaningful deviations from national league boundaries. Examples include the persistent integration of the Premier League and Championship

into a single transfer ecosystem and the temporary merging of Bundesliga and Eredivisie clubs during the 2020/2021 season, suggesting that transfer activity can sometimes create cross-league structures that are not captured by league membership alone.

The robustness analysis further demonstrated that the detected communities are not an artifact of a small number of transfers or clubs. The overall partition remained remarkably stable under random edge removal, hub removal, and variations in the transfer threshold, indicating that the observed community structure reflects genuine properties of the transfer market.

Finally, the temporal analysis revealed that the community structure remains relatively stable across seasons. Similar values of modularity, NMI, ARI, and community purity were obtained throughout the 2018/2019–2022/2023 period, while several league-centered communities were consistently recovered over time. This suggests that the main structural patterns of the European transfer market persist despite the continuous movement of players and clubs.

Overall, our findings indicate that community detection provides an effective framework for analysing football transfer networks and recovering meaningful organizational structures from transfer activity alone. Future work could extend this study by incorporating transfer fees, analysing directed transfer flows, or developing more sophisticated methods to explicitly track the evolution of individual communities across seasons.

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Appendix

Header	Description	Data Type
<code>club_name</code>	name of club	text
<code>player_name</code>	name of player	text
<code>position</code>	position of player	text
<code>club_involved_name</code>	name of secondary club involved in transfer	text
<code>fee</code>	raw transfer fee information	text
<code>transfer_movement</code>	transfer into club or out of club?	text
<code>transfer_period</code>	transfer window (summer or winter)	text
<code>fee_cleaned</code>	numeric transformation of <code>fee</code> , in EUR millions	numeric
<code>league_name</code>	name of league <code>club_name</code> belongs to	text
<code>year</code>	year of transfer	text
<code>season</code>	season of transfer (interpolated from <code>year</code>)	text
<code>country</code>	country of league	text

Figure 1: Variables available in the transfer datasets.

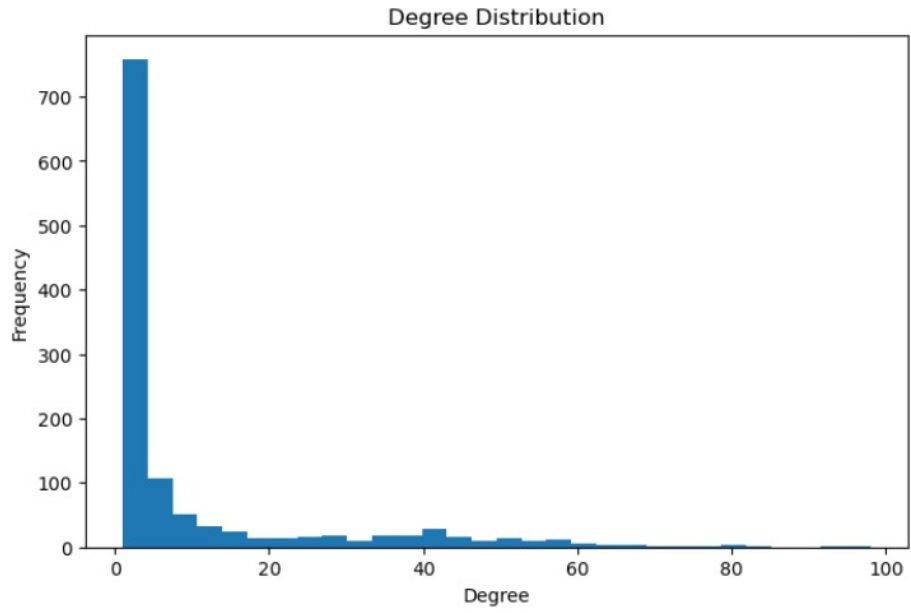


Figure 2: Degree distribution of the transfer network for the network comprising seasons 2020/2021 to 2022/2023.

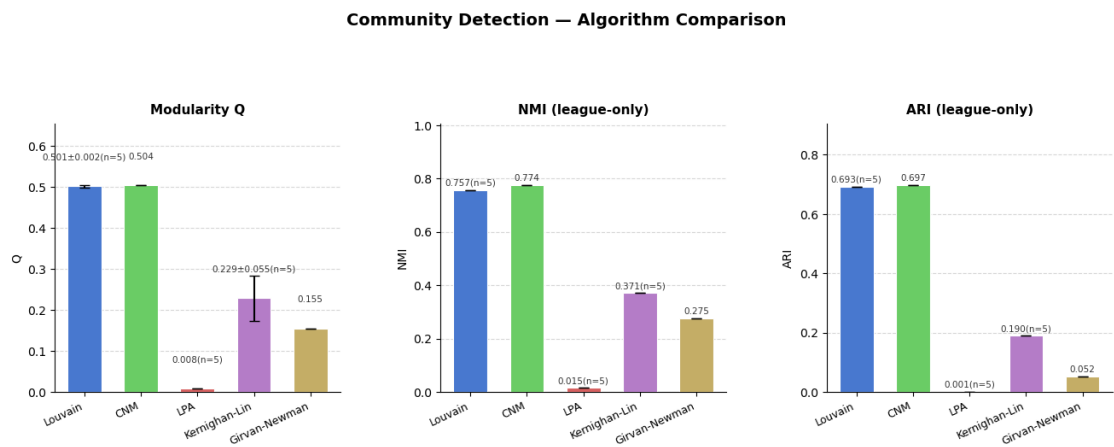


Figure 3: Final scores of all algorithms

Louvain Communities

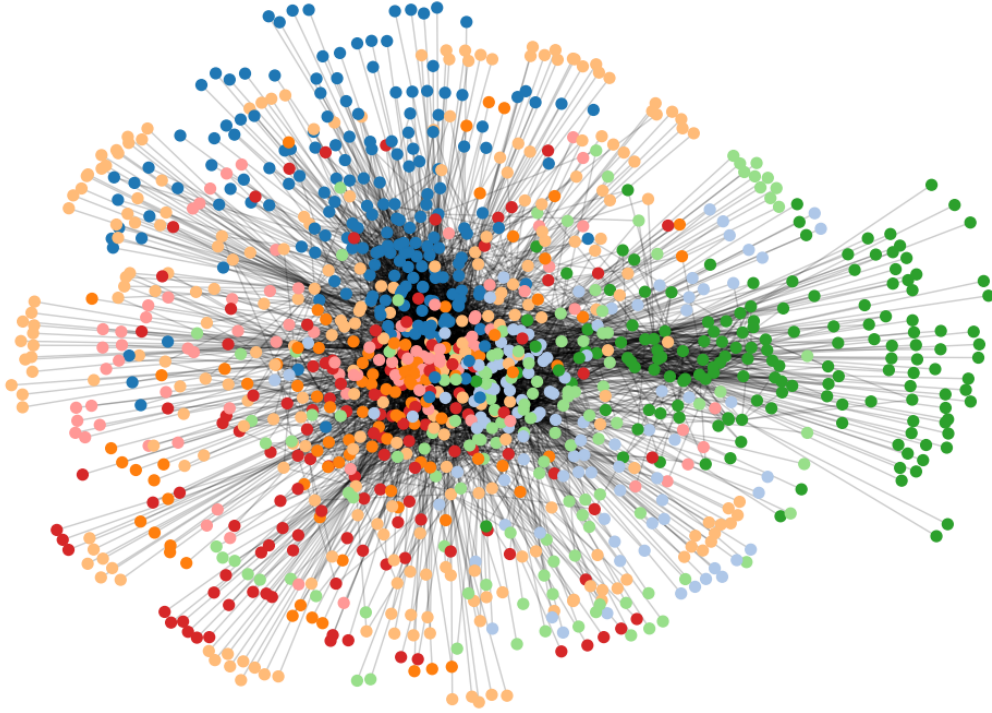


Figure 4: Graph of the Network comprising seasons 2020/2021 to 2022/2023 with its detected communities by the Louvain algorithm.

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COMMUNITY 2
Number of clubs: 129
["1. FC Köln", "1. FC Heidenheim", "1. FC K'lauren", "1. FC Magdeburg", "1. FC Nuremberg", "1. FC Union Berlin", "A. Bielefeld", "A. Klagenfurt", "ADO", "AZ", "Aarhus GF", "Aberdeen FC", "Ajax Amsterdam", "Alianza Atl.", "Almere City", "Anderlecht", "Ankaragücü", "Antwerp", "Austria Vienna", "BSC Young Boys", "Bay. Leverkusen", "Bayer 04 Leverkusen", "Bayern Munich", "Bogotá FC", "Bohemians", "Borussia Dortmund", "CA Atenas", "CD Cruz Azul", "Cambuur", "Cercle Brugge", "Club Bruges", "Club NXT", "Copenhagen", "Darmstadt", "De Graafschap", "Den Bosch", "Dinamo", "Dordrecht", "Dynamo Dresden", "Düsseldorf", "Eintracht Frankfurt", "Emmen", "Erzgebirge Aue", "Estudiantes", "Ettifaq FC", "Excelsior", "FC Augsburg", "FC Flora", "FC Groningen", "FC Ingolstadt", "FC Schalke 04", "FC St. Pauli", "FC Twente", "FC Utrecht", "FC Volendam", "Feyenoord", "Fortuna", "Fortuna Sittard", "Gaziantep FK", "Go Ahead", "Godoy Cruz", "Greuther Fürth", "HJK Helsinki", "HNK Gorica", "Hallescher FC", "Hamburg", "Hannover 96", "Heerenveen", "Hertha BSC", "Hoffenheim", "Jahn Regensburg", "Júbilo Iwata", "K. Offenbach", "KSK Beveren", "KSK Lierse Kem.", "KV Mechelen", "Kansas City", "Karagümrük", "Karlsruher SC", "Koninklijke HFC", "KuPS", "Laranja (PR)", "Lech Poznań", "Legia Warszawa", "MAS Fes", "MVV", "Montréal", "NK Lokomotiva", "NK Osijek", "Nordsjaelland", "OFI Crete FC", "Oberhausen", "PAS Giannina", "PEC Zwolle", "PSV", "Panathinaikos", "Philadelphia", "Podbeskidzie", "RB Leipzig", "RKC", "RSCA Futures", "Rapid Vienna", "Roda JC", "SC Freiburg", "SC Verl", "SCR Altach", "SKN St. Pölten", "SV Werder Bremen", "Salt Lake", "San Jose", "Sheff Utd", "Stabæk Fotball", "Stormv. Telstar", "Strømsgodset", "Stuttgart", "TOP", "Trabzonspor", "Turgutluspor", "Twente Enschede FC", "VCU Rams", "VfL Bochum", "VfL Wolfsburg", "Viktoria Köln", "Vitesse", "WSG Tirol", "Wehen Wiesbaden", "Willem", "Without Club", "Örebro SK"]

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Figure 5: Detail of the list of teams contained inside the community n^o2 for the Clauset–Newman–Moore algorithm.

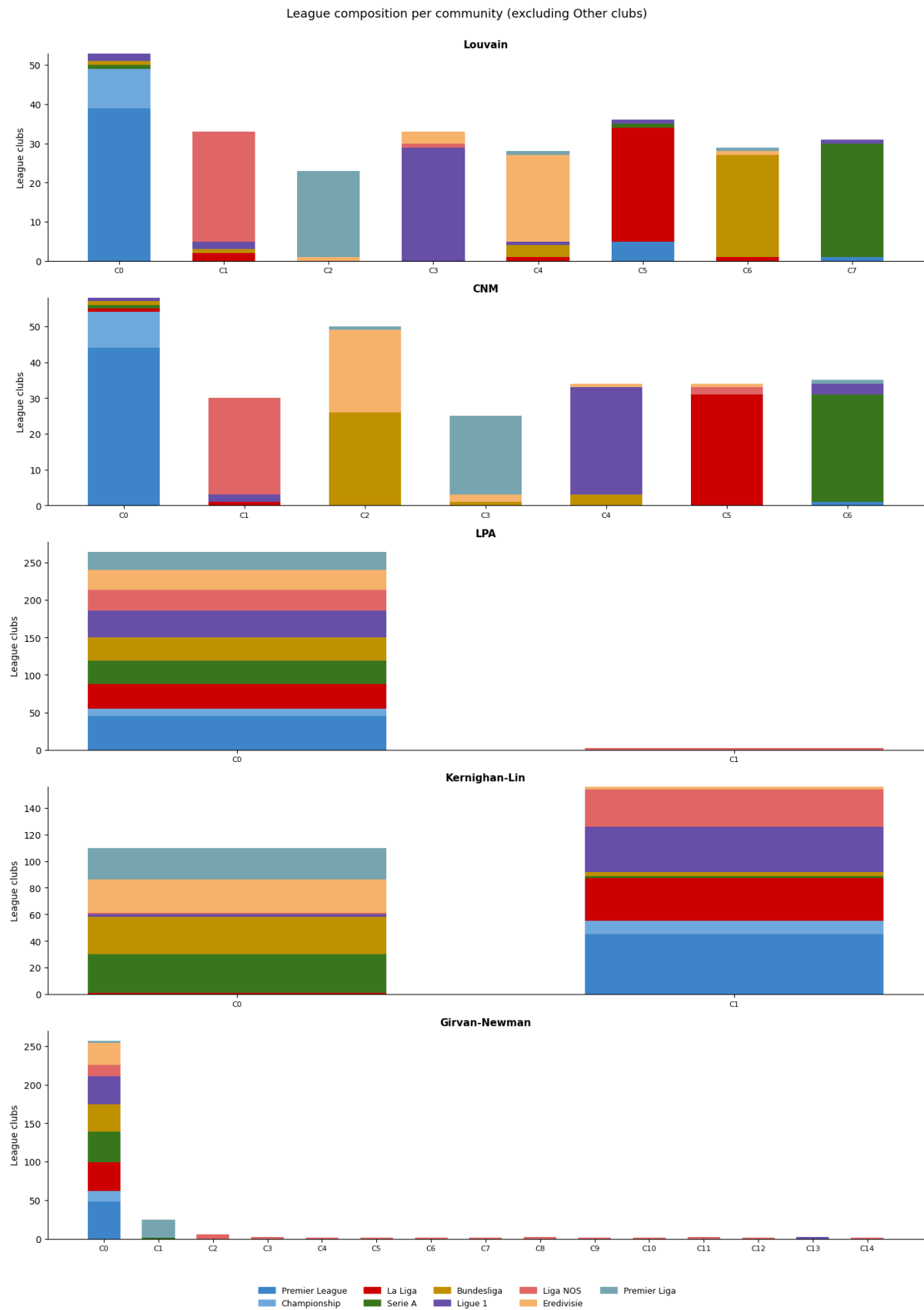


Figure 6: Leagues distribution into communities

Robustness Analysis of Community Structure

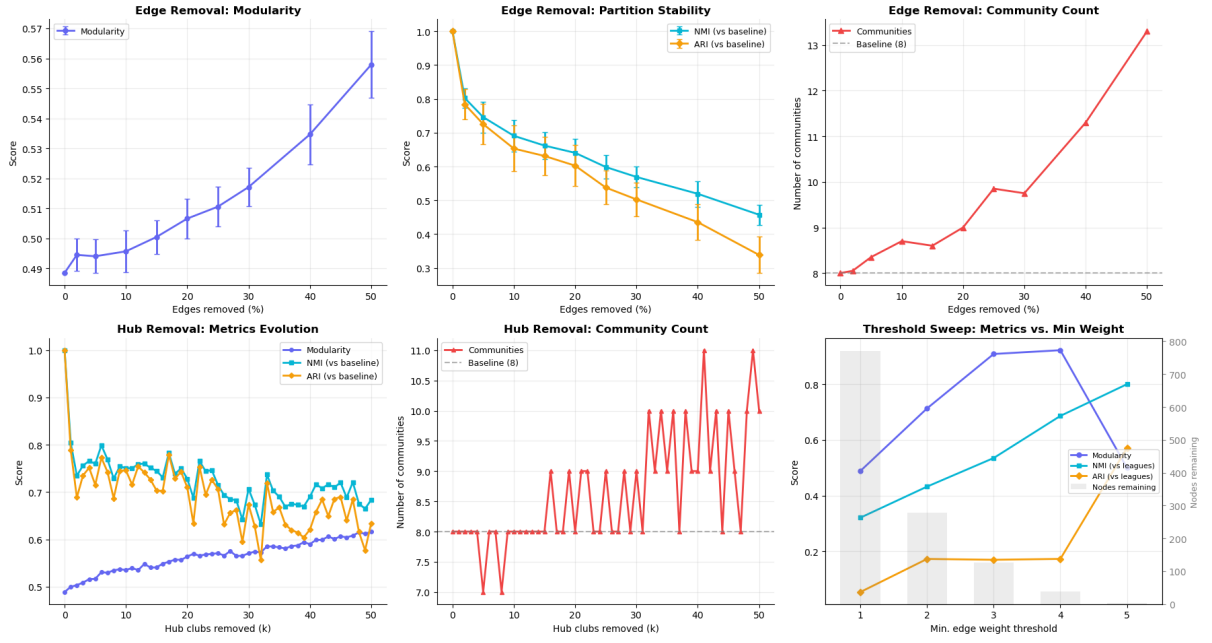


Figure 7: Outcomes of the experiments about robustness

A Data Loading and Preprocessing

The data used in this study is obtained from the following public repository: <https://github.com/ewenme/transfers/tree/master>

The repository contains player transfer records scraped from Transfermarkt and made freely available to encourage research and analysis in football analytics. The data is provided as CSV files, with each file corresponding to a different European national league. The leagues considered in this dataset are:

- English Premier League (`premier-league.csv`)
- English Championship (`championship.csv`)
- French Ligue 1 (`ligue-1.csv`)
- German Bundesliga (`1-bundesliga.csv`)
- Italian Serie A (`serie-a.csv`)
- Spanish La Liga (`primera-division.csv`)
- Portuguese Liga NOS (`liga-nos.csv`)
- Dutch Eredivisie (`eredivisie.csv`)
- Russian Premier Liga (`premier-liga.csv`)

Each dataset follows a common schema. The variables available in each dataset are shown in Figure 1 in the appendix.

The first stage of the study consisted of an exploratory analysis of the raw datasets. We began by inspecting the format and meaning of each variable and examining individual transfer records in detail. This process revealed that each transfer was recorded twice in the data: once as an outgoing transfer in the dataset corresponding to the source club’s league, and once as an incoming transfer in the dataset corresponding to the destination club’s league. To avoid duplicate records, all league datasets were merged into a single dataset and only transfers labelled as *in* were retained.

The exploratory analysis also revealed several data quality challenges that required a substantial data cleaning and engineering effort. In particular, the same club could appear under different names across leagues and seasons. For example, a club recorded as *Wolverhampton Wanderers* in one dataset could appear as *Wolves* in another. Similarly, transfers involving youth teams were sometimes recorded using names such as *Manchester United U19*, creating additional inconsistencies.

To address these issues, we implemented a club name normalization process combining fuzzy string matching techniques with manually curated mapping dictionaries for the most common naming variations identified during exploration. This proved to be a delicate and time-consuming task, representing a significant part of the overall data preparation process. Nevertheless, it was essential to ensure the highest possible data quality before constructing the transfer network.

Another challenge concerned clubs belonging to leagues not explicitly included in our dataset. While the transfer records focus on the selected European leagues, many transfers involve clubs from external leagues outside the scope of the study. To account for these cases, all such clubs were assigned to a separate league category labelled *Other*. This allowed them to be incorporated into the network while preserving a meaningful ground-truth league assignment for later evaluation of the detected communities.

B AI Statement

Generative AI tools (ChatGPT, OpenAI) were used as programming assistants during selected stages of the project. In particular, AI-assisted code generation was used to support the data

preprocessing pipeline, including the identification of clubs with similar names through fuzzy matching techniques and the development of regular-expression-based procedures for detecting naming patterns and inconsistencies across datasets.

AI assistance was also used to help implement parts of the temporal community evolution analysis by adapting and extending the code structure already developed for the other experiments. All generated code was reviewed, modified when necessary, and validated by the authors before being incorporated into the final code.

C Division of work

The project was initially undertaken by a group of three students. However, one member withdrew from the course during the semester and did not participate in the completion of the project. As a result, all remaining work was carried out jointly by Marc and Alessandro.

The division of responsibilities was organised as follows, both for the implementation of the experiments and for the corresponding sections of the written report:

- **Marc:** data preprocessing and preparation, club-name normalization, Louvain community detection experiments, and temporal community-evolution analysis.
- **Alessandro:** comparative evaluation of community detection algorithms and robustness/sensitivity analysis experiments.

Although responsibilities were divided according to these primary areas, both authors contributed to the overall design of the study, discussion of results, revision of the codebase, and final editing of the report.